TASK SPECIFICATION: P5-01 PHASE P5 — SMART PARKING & ALPR COMPUTER VISION GATE

Parking Domain Schema (Zones, Spots, Assignments, Tariffs)

Git Branch: `feature/P5-01-parking-schema` · Target Pull Request: feature/P5-01-parking-schema

Core Objective & Technical Scope:
Database schema for campus parking zones, 30 monitored spots, vehicle permits, and dynamic hourly parking tariff schedules.

Priority Level	High	Complexity Rating	`M`
Execution Status	Backlog 📋	Target PR / Commit	`feature/P5-01-parking-schema`
Upstream Dependencies	P2-01		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Author V6__parking.sql migration creating parking_zone, parking_spot, and tariff tables.
2. Create Zone, Spot, and Tariff JPA entities and repositories.
3. Seed baseline definitions for 30 monitored campus spots across North and South zones.
4. Validate foreign keys and constraints.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `backend/src/main/resources/db/migration/V6__parking.sql`
- `parking/domain/*`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Spot numbering unique per zone.
- Tariff schedules support dynamic hourly rates and faculty exemption rules.

VERIFICATION & TEST COVERAGE STRATEGY

- ParkingRepositoryTest
- Flyway migration test